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Vaishak Prasad

Curriculum Vitae – Gravitational Wave Astronomy and Numerical Relativity

Positions held

Jan 25 - Present **Physics At the eXtreme (PAX) group, Institute for Gravitation and the Cosmos (IGC), The Pennsylvania State University (PSU)**,
Eberly Postdoctoral Scholar and co-lead of the Cosmic Explorer Science Traceability Matrix - Testing General Relativity, With: Prof. B. S. Sathyaprakash.

Mar 23 - Dec 24 **International Centre for Theoretical Sciences - Tata Institute of Fundamental Research (ICTS-TIFR), Bengaluru, India**,
Post-doctoral fellow, With: Dr. Prayush Kumar.

Nov 22 - Mar 23 **International Centre for Theoretical Sciences - Tata Institute of Fundamental Research (ICTS-TIFR), Bengaluru, India**,
Visiting student, Host: Dr. Prayush Kumar.

2016–22 **Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India**,
Senior Research Scholar and Shyama Prasad Mukherjee Fellow (CSIR).
PhD Supervisor: Prof. Sukanta Bose

Ph.D. Thesis

Thesis Title Strong Gravity Physics of Dynamical Horizons in Black Hole Mergers and its Imprint in their Gravitational Radiation

Supervisor **Prof. Sukanta Bose**.

Defended **29 March 2023**.

Degree awarded **6 April 2023**.
Jawaharlal Nehru University, New Delhi (through IUCAA, Pune)

Education

2012–2016 **Birla Institute of Technology and Science, Pilani (BITS Pilani) - Hyderabad Campus**,
Physics, Integrated Master of Science (honors), GPA – 8.71/10, Physics GPA – 9.36/10.
M.Sc. Thesis: The Causal Structure of Kerr-Newman Spacetime. Supervisor: Dr. Sashideep Gutti.

Research Interests

Physics Dynamical horizons and strong-field black hole dynamics. Numerical relativity. Tests of general relativity. Black hole spectroscopy and ringdown. Binary black hole population inference. Third-generation and space-based detector science (Cosmic Explorer, Einstein Telescope, LISA).

Methods Gravitational wave data analysis and Bayesian inference. Gravitational wave searches and template banks. Waveform modelling and surrogates. High-performance computing. GPU computing. Machine learning.

Research and Work Experience

2018 - **Dynamical horizons and strong-field black hole dynamics.** Quasi-local horizon geometry in binary mergers. Horizon multipoles, shear and tidal deformability. The universal structure of horizon formation. The quantitative area law and balance-law violation.

2023 - **Tests of general relativity and black hole spectroscopy.** GR-informed ringdown modelling. Multi-segment consistency tests. Quasi-normal mode instabilities. The apparent GR violation in GW230814. Maximum entropy in mergers.

2025 - **Gravitational wave data analysis and inference.** Exact accelerated time-domain likelihoods. Batched likelihood evaluation for nested sampling. Performance and convergence of stochastic samplers. Spin priors and population inference.

2023 - **Waveform modelling, surrogates and template banks.** Spectral numerical relativity surrogates with analytic derivatives. Massively parallel higher-order-mode template banks for precessing binaries. Eccentric and precessing binary simulations.

2018 - **Numerical relativity methods and infrastructure.** Cauchy-characteristic matching on compactified null slices. Pseudo-spectral methods and numerical PDEs. Apparent horizon finding in SpEC and SpECTRE. Compile-time optimization and HPC benchmarking. Comparison infrastructure for NR catalogues.

2025 - **Next-generation detector science.** Science traceability and tests of general relativity for Cosmic Explorer. Forecasting for third-generation and space-based detectors.

2018 - 2021 **Scientific computing and visualization.** Parallel computing and data I/O on HPC systems. 3D scientific visualization of numerical relativity data.

2015 - 2020 **Earlier work.** Causal structure of spacetime, cosmic censorship and chronology protection. Black holes in higher dimensions. Scattering of ringdown signals from black holes surrounded by matter. Magnetohydrodynamics and galactic dynamos. Cosmological parameter estimation from supernovae.

Publications

Published and archived

19. **Vaishak Prasad**, *Prayush Kumar, Michael Boyle, Lawrence E. Kidder, Harald P. Pfeiffer, Mark A. Scheel*, "A catalog of long, eccentric and precessing binary black hole simulations from ICTS-TIFR", arXiv preprint, 14 October 2026, gr-qc:2610.19688.

18. Yu-Cun Xie, ***Vaishak Prasad***, "Universal Structure of Horizon Formation in Generic Binary Black Hole Mergers", arXiv preprint, 10 September 2026, gr-qc:2609.11819.

17. **Vaishak Prasad**, "NRHJSur3dq8: a spectral numerical-relativity surrogate family for non-eccentric aligned-spin binary-black-hole waveforms", arXiv preprint, 8 September 2026, gr-qc:2609.09088.
    Single author. Released with the public `nrhjsurrogate` package.

16. **Vaishak Prasad**, *Monica Rincón-Ramírez, Nathan K. Johnson-McDaniel, Eugenio Bianchi, Ish Gupta and B. S. Sathyaprakash*, "Do black hole mergers maximize entropy?", Published 27 August 2026, Int. J. Mod. Phys. D, 2641013..
    Gravity Research Foundation 2026 essay, Honourable Mention.

15. Monica Rincón-Ramírez, *Nathan K. Johnson-McDaniel, Eugenio Bianchi, Ish Gupta,* ***Vaishak Prasad*** *and B. S. Sathyaprakash*, "Maximum Entropy Conjecture for Black Hole Mergers", Published 10 July 2026, Phys. Rev. Lett. 137, 021406., arXiv:2601.22388.
    Devised the numerical relativity method used to test the conjecture. The conjecture is framed in terms of the Bondi mass and angular momentum. Also contributed the horizon-geometry analysis.

14. **Vaishak Prasad**, "Accelerated Time-domain Analysis for Gravitational Wave Astronomy", arXiv preprint, 6 March 2026, gr-qc:2603.06010.
    Single author. Exact time-domain likelihood evaluation at near-FFT cost via the Gohberg–Semencul representation. Released with the `tdanalysis` package.

13. **Vaishak Prasad**, "Multi-Segment Consistency Tests of General Relativity", arXiv preprint, 6 March 2026, gr-qc:2603.05835.
    Single author. Demonstrated on GW250114. To be proposed as a reviewed LVK test of general relativity.

12. **Vaishak Prasad**, *and B. S. Sathyaprakash*, "Are black hole spins truly near-zero?", arXiv preprint, 13 February 2026, gr-qc:2602.12859.
11. Divyajyoti, *Isobel M. Romero-Shaw,* ***Vaishak Prasad****, Kaushik Paul, Chandra Kant Mishra, Prayush Kumar, Akash Maurya, Michael Boyle, Lawrence E. Kidder, Harald P. Pfeiffer, Mark A. Scheel*, "Biased parameter inference of eccentric, spin-precessing binary black holes", Published 12 May 2026, Phys. Rev. D 113, 103022., arXiv:2510.04332.
10. **LVK collaboration**, *incl.* ***Vaishak Prasad***, "Black Hole Spectroscopy and Tests of General Relativity with GW250114", Published 30 January 2026, Phys. Rev. Lett. 136, 041403., arXiv:2509.08099.
    Contributed to the analysis and interpretation of the area-law result and its connection with the consistency test. Also contributed to the ringdown model and the review of results. Editors' Suggestion. Featured in *Physics*.
9. **LVK collaboration**, *incl.* ***Vaishak Prasad***, "GW250114: Testing Hawking's Area Law and the Kerr Nature of Black Holes", Published 10 September 2025, Phys. Rev. Lett. 135, 111403..
    Contributed to the analysis and interpretation of the area-law result and to the ringdown model.
8. **LVK collaboration**, *incl.* ***Vaishak Prasad***, "GW230814: investigation of a loud gravitational-wave signal observed with a single detector", Published 18 June 2026, Astrophys. J. Lett. 1004, L23., arXiv:2509.07348.
    Provided the targeted numerical relativity simulation and contributed to writing the paper.
7. **Vaishak Prasad**, "The Shear at the Common Dynamical Horizon in Binary Black Hole Mergers and its Imprint in their Gravitational Radiation.", Published 28 April 2025, Phys. Rev. D 111, 084070..
6. P. Ajith et al., *incl.* ***Vaishak Prasad***, "The Lunar Gravitational-wave Antenna: Mission Studies and Science Case.", Published 28 January 2025, JCAP 01 (2025) 108..
    Author of the Testing GR section
5. **Vaishak Prasad**, "Tidal deformation of dynamical horizons in binary black hole mergers and its imprint on gravitational radiation.", Published 16 February 2024, Phys. Rev. D 109, 044033., arXiv:2109.01193.
    First to show that the horizons of a binary acquire multipolar deformations (hair) during the inspiral, strongly correlated with the gravitational-wave signal at infinity, which the common horizon inherits at merger and loses by absorbing radiation, with quasinormal decay of its multipole moments (damping rate of the dominant moment within 3% of the quasinormal value).
4. Sashideep Gutti, *Shailesh Kulkarni,* ***Vaishak Prasad***, "Closed timelike curves and energy conditions in regular spacetimes", Published 15 December 2022, Eur. Phys. J. C (2022) 82: 1136..
    Equal contribution
3. **Vaishak Prasad**, *Anshu Gupta, Sukanta Bose, Badri Krishnan*, "Tidal deformation of dynamical horizons in binary black hole mergers.", Published 9 February 2022, Phys. Rev. D 105, 044019., arXiv:2106.02595.
2. **Vaishak Prasad**, *Anshu Gupta, Sukanta Bose, Badri Krishnan, and Erik Schnetter*, "News from Horizons in Binary Black Hole Mergers", Published 16 September 2020, Phys. Rev. Lett. 125, 121101., arXiv:2003.06215.
1. **Vaishak Prasad**, *Rahul Srinivasan, Sashideep Gutti*, "The Chronology Protection Problem in modified Kerr-Newman space-times.", Published 14 January 2019, Phys. Rev. D 99, 024023., arXiv:1803.06666.

Theses and reports

2023 Strong Gravity Physics of Dynamical Horizons in Black Hole Mergers and its Imprint in their Gravitational Radiation. Ph.D. thesis, Jawaharlal Nehru University (IUCAA, Pune). Supervisor: Prof. Sukanta Bose. Submitted version in Shodhganga.

2017 Large Scale Magnetic Fields and the Turbulent Dynamo. Project report, IUCAA-NCRA Graduate School. Supervisor: Prof. Kandaswamy Subramanian.

2014 Turbulent Diffusion. Practice School report, BITS Pilani, at IUCAA, Pune.

In preparation

10. **Vaishak Prasad**, *B. S. Sathyaprakash and Abhay Ashtekar*, "Horizon-area increase in GW250114: the quantitative area law test"., In preparation.
9. **Vaishak Prasad**, *Abhay Ashtekar and B. S. Sathyaprakash*, "Assessing intrinsic waveform accuracy using the waveform balance laws"., In preparation.

8. **Vaishak Prasad**, "Waveform systematic measurements using the multi-segment consistency test"., LIGO-P2600044-v1, in preparation.
7. **Vaishak Prasad**, "Parallel Template Bank Generation for Non-Eccentric Precessing Binaries"., LIGO-P2600567, in LIGO-Virgo-KAGRA internal review (P&P).
6. **Vaishak Prasad**, *B. S. Sathyaprakash and Abhay Ashtekar*, "A universal law for the balance-law violation of finite-length gravitational waveforms: post-Newtonian origin and the geometry of precession"..
5. Viviana Caceres, ***Vaishak Prasad***, "Time-domain black-hole spectroscopy with a segmented, GPU-accelerated likelihood"..
4. **Vaishak Prasad**, "Cauchy-characteristic matching via generalized-harmonic evolution on compactified null slices. I. Formulation and the hyperbolicity of exact null slicing"..
3. **Vaishak Prasad**, "Batched likelihood evaluation for nested sampling in gravitational-wave parameter estimation"..
2. **Vaishak Prasad**, "Compile-time performance optimization of numerical relativity simulations in C++ using SpEC"..
1. Priyanka Sinha, *Samanwaya Mukherjee, Prayush Kumar,* ***Vaishak Prasad****, Lawrence Kidder, Harald Pfeiffer, Mark Scheel, Md Shaikh, Michael Boyle*, "Can we twist gravitational waveforms from planar eccentric binaries to emulate generic ones"..

Fellowships, Grants, and Awards

Pending

Oct 2026 **Principal Investigator, ERC Starting Grant** (ERC-2027-StG, panel PE9), *HORIZONS – Signatures of dynamical geometry: observables of dynamical horizons and precision parameter estimation with third-generation detectors*. Five years, EUR 1,771,198 requested. Host: Radboud University, Nijmegen, The Netherlands.

Sep 2026 **Co-PI (shared with B. S. Sathyaprakash), CHiwi Foundation**: *The Enigma of Black Holes – From Horizons to Gravitational Waves*. A two-year programme connecting the multipolar structure of newly formed black hole horizons to the quasi-normal mode content of the radiation observed at infinity, with forecasts for Cosmic Explorer, the Einstein Telescope and LISA. Pennsylvania State University.

Sep 2026 **Applicant, Marie Skłodowska-Curie Actions Postdoctoral Fellowship** (MSCA-PF-2026), *TIMBRE*: time-domain inference for the LISA global fit. Host: University of Birmingham, UK, with a secondment to the Max Planck Institute for Gravitational Physics (AEI), Potsdam.

Aug 2026 **Applicant, NWO Talent Programme Veni** (2026 round): Strong-field gravity at the horizon – observables, quasi-local diagnostics, and accuracy requirements for next-generation numerical relativity. Host: Radboud University, Nijmegen, The Netherlands.

Awarded

May 2026 **Honourable Mention, Gravity Research Foundation Awards for Essays on Gravitation 2026**, for *Do black hole mergers maximize entropy?*

Aug 2026 - Jul 2029 **co-Principal Investigator, NSF award PHY-2608011**, *Collaborative Research: Testing General Relativity with Gravitational Wave Observations* (USD 360,000, 1 August 2026 to 31 July 2029, PI Prof. B. S. Sathyaprakash). Conceived and authored the funded component on tests of general relativity through ringdown inference and spectroscopy informed by numerical relativity.

2026 **NSF ACCESS computing allocations** PHY260122, PHY260124 and PHY260295.

Nov 2025 **APS Early Career Travel Award, MAS 2025** for a presentation at the APS Mid-Atlantic Section - Harrisburg 2025.

Sep 2025 - Sep 2026 **Principal Investigator, NSF ACCESS Explore Grant** PHY250294, *Accelerated time-domain analysis for gravitational wave astronomy*.

Jan 2025 - **Eberly Post-Doctoral Fellowship**, Physics at the eXtreme (PAX) group, Institute for Gravitation and the Cosmos (IGC), Pennsylvania State University (PSU), PA, USA.

Mar 2023 - Dec 2024 **Post-Doctoral Fellowship**, International Centre for Theoretical Sciences - Tata Institute of Fundamental Research (ICTS-TIFR), Bengaluru, India.

Jan 2018 - 2022 **Shyama Prasad Mukherjee Fellowship**, Council of Scientific and Industrial Research (CSIR), Govt. of India.

June 2016 **Junior Research Fellowship (JRF) offered at**.

- Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India.
- Tata Institute of Fundamental Research (TIFR), Mumbai, India.
- Indian Institute of Science (IISc), Bengaluru, India.

Accepted the JRF position at IUCAA.

May-Jul 2015 **G N Ramachandran Fellowship For Young Promising Researchers**, Indian Institute of Technology Madras (IITM).

May-Jul 2015 **Summer Research Fellowship**, Indian Institute of Technology Madras (IITM).

Aug 2013 Fellow, **Kishore Vaigyanik Protsahan Yojana** (KVPY), Department of Science and Technology (DST), Govt. of India.

Feb 2013 Institute Semester Merit Scholarship, BITS Pilani - Hyderabad Campus.

Achievements

Dec 2016 All India Rank 4, National Eligibility Test (UGC-CSIR NET).

Mar 2016 All India Rank 6, Joint Entrance Screening Test (JEST).

Invited Talks and Seminars

Aug 2026 Goethe University Frankfurt, Germany. ***On the Role of Numerical Relativity in Testing GR using LIGO-Virgo-KAGRA Observations of Binary Black Hole Mergers***

Jul 2026 School of Physics and Astronomy, University of Nottingham, UK. ***Precision Strong-Field Gravity with Gravitational Waves: Consistency Tests, Black-Hole Spins, and Exceptional Events***

Jul 2026 Scuola Internazionale Superiore di Studi Avanzati (SISSA), Trieste, Italy. ***Precision Strong-Field Gravity with Gravitational Waves: Consistency Tests, Black-Hole Spins, and Exceptional Events***

Jul 2026 Gran Sasso Science Institute (GSSI), L'Aquila, Italy. ***Precision Strong-Field Gravity with Gravitational Waves: Black-Hole Spins, Waveform Systematics, and Exceptional Events***

Jul 2026 Theoretical Physics Institute, Friedrich Schiller University Jena, Germany. Two research seminars: ***On the Role of Numerical Relativity in Testing GR using LIGO-Virgo-KAGRA Observations of Binary Black Hole Mergers*** and ***Strong field dynamics of dynamical horizons in binary mergers***

Jun 2026 Astrophysical and Cosmological Relativity division, Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Potsdam, Germany. ***Precision Parameter Estimation, Strong Field Tests of GR, and Spin Inference for Future Detections***

Jun 2026 Gravitational Physics group, School of Physics and Astronomy, Cardiff University, UK. ***Parameter Estimation and Precision Tests of GR for Next-Generation Gravitational Wave Detections***

Jun 2026 Department of Physics and Astronomy, Washington State University, Pullman, WA (online seminar). ***Parameter Estimation and Precision Tests of GR for Next-Generation Gravitational Wave Detections***

May 2026 Institute for Gravitational Research, University of Glasgow, UK. ***Precision Parameter Estimation, Strong Field Tests of GR, and Spin Inference for Future Detections***

Mar 2026 Strong Gravity Seminar, Institute for Gravitation and the Cosmos, Pennsylvania State University. ***Testing Black Holes in General Relativity Using their Horizon Areas***

Mar 2026 Beginner's Guide to Modeling for Gravitational Wave Detection, APS Global Physics Summit 2026, Denver, CO. ***Numerical Relativity in Gravitational-Wave Astronomy: From Waveforms to Horizons***

Mar 2026 Gravitational Physics Seminar Series, IIT Gandhinagar and IACS Kolkata, India. ***A Maximum Entropy Conjecture for Binary Black Hole Mergers***

Jun 2025 NP3M Summer School, Indiana University, IN. Invited lecture on ***Introduction to stochastic sampling***

Mar 2025 APS DGRAV Seminar Series. ***Strong field physics of dynamical horizons in binary black hole mergers and its imprint in their gravitational radiation***

Jul 2024 Physics at the eXtreme (PAX) Group, Pennsylvania State University. ***Dynamical Horizons and Numerical Relativity: Strong correlations***

Mar 2024 Numerical and Analytical Relativity (NAR-2024). ***Physics of Dynamical Horizons and their Applications to Numerical Relativity***

Mar 2024 Centre for Strings, Gravitation and Cosmology (CSGC), IIT Madras, Chennai, India. ***Tidal deformation of dynamical horizons in binary black hole mergers and its imprint in their gravitational radiation***

Feb 2024 Dept. of Physics, IIT Bombay, Mumbai, India. ***Tidal deformation of dynamical horizons in binary black hole mergers and its imprint in their gravitational radiation***

Feb 2024 Gravity wave group, IIT Madras, Chennai, India. Invited lecture on ***Numerical Relativity***

Feb 2022 Second Chennai Symposium on Gravitation and Cosmology (Virtual). ***Tidal deformability of dynamical horizons in binary black hole mergers***

Contributed Talks and Posters

Jun 2026 Eastern Gravity Meeting 2026, Cornell University, Ithaca, NY. Contributed talk on ***Multi-segment consistency tests of General Relativity***

May 2025 Science Traceability Matrix for Cosmic Explorer, Penn State, State College, PA. Talk on ***Overview of Science Traceability Matrix - Tests of General Relativity***

Apr 2025 APS Global Summit, Anaheim, CA. Talk on ***News from dynamical horizons in binary black hole mergers***

Aug 2024 Black Holes Inside Out, NBI, Copenhagen. Poster on ***News from Horizons in BBH mergers***

Aug 2024 Ringdown Inside Out, NBI, Copenhagen. Contributory discussion on ***Quasi-normal ringing at the common dynamical horizons in BBH mergers***

Aug 2024 SXSCon2024, ICERM. SpEC Developer session on ***Compile-time optimization of SpEC simulations: How to make the best of HPCs***

Aug 2024 SXSCon2024, ICERM. Lightning talk on ***Applications of dynamical horizons in SpEC and SpECTRE***

Dec 2023 International Conference on Gravitation and Cosmology (ICGC), IIT Guwahati. Talk on ***Tidal deformation of dynamical horizons in binary black hole mergers and its imprint in their gravitational radiation***

Jul 2021 Amaldi 14. Poster on ***Tidal deformability of dynamical horizons in binary black hole mergers***

Dec 2019 International Conference on Gravitation and Cosmology (ICGC), IISER Mohali. Talk on ***The relation between horizon shear and gravitational waves at null infinity***

Jan 2019 30th meeting of the Indian Association for General Relativity and Gravitation (IAGRG), Hyderabad. Talk on ***Chronology protection problem in modified Kerr-Newman spacetimes***

Jul 2018 Les Houches Summer School on Gravitational Waves, France. Talk on ***The Chronology protection problem in modified Kerr-Newman spacetimes***

Seminars at IUCAA

Jul 2022 Ph.D. submission seminar. ***Strong Gravity Physics of Dynamical Horizons in Black Hole Mergers and their Gravitational Radiation***

Nov 2021 A seminar on ***Tidal Deformability of Dynamical Horizons in Binary Black Hole Mergers*** in the IUCAA student seminars.

Feb 2020 A seminar on ***The relation between the shear and News in binary black hole mergers*** in the IUCAA student seminars.

Jul 2018 Talk on ***The Chronology Protection Problem*** in the monthly IUCAA last Friday talks.

Jun 2018 A seminar on ***Dynamical Black holes*** in the IUCAA Academic retreat.

2018 Talk on ***Dynamical horizons*** in the IUCAA Gravitational Wave Research Group.

2018 Led a series of talks and discussions on ***Isolated and Dynamical Horizons*** for members of the IUCAA Gravitational Wave Research Group.

Projects Guided, Mentoring and Teaching

Supervision and mentoring

2025 - Co-mentoring five graduate students in the Physics at the eXtreme (PAX) group, Penn State, on their research projects.

2025 - **Nathaniel Kirby** and **Sanika Khadkikar**, graduate students of Prof. B. S. Sathyaprakash, PAX Group, Penn State. Co-mentored on projects including the reanalysis of catalogued gravitational-wave events with geometrically motivated spin priors.

Aug-Sep 2025 **Giovanni Benetti**, co-tutored Master's project on *Impact of waveform systematics on the sky-localization of gravitational wave sources*, PAX Group, Penn State.

Jul-Aug 2025 **Alexander Torres**, supervised NSF REU project on *Modeling the merger time of binary black holes using Machine Learning and Numerical Relativity Data*, PAX Group, Penn State.

2023 - **Bhuvaneshwari Kashi**, co-supervised visualizations from numerical relativity simulations of black hole mergers, ICTS-TIFR, Bengaluru.

2023 **Arushi Kolluru**, co-supervised project on gravitational waves from binary black hole mergers, ICTS-TIFR, Bengaluru.

2019-20 **Sanjeet Sahasrabudhe**. Supervised the Bachelor's thesis on *Numerical methods for solving classical dynamical systems and their application to the precession of the orbits*. Link to the thesis. Currently working as a Data Scientist.

Teaching

Fall 2026 Taught a few lectures of the graduate course on **general relativity** (Prof. B. S. Sathyaprakash), Pennsylvania State University. Assessed students' assignments.

Jun 2025 Lecturer on stochastic sampling and tutor for the course on the Geometric Approach to Gravitational Wave Data Analysis (Prof. B. S. Sathyaprakash), **NP3M Summer School 2025**, Indiana University, IN, USA.

Jul 2024 Tutor for the course on ultralight boson clouds around black holes (Dr. Nils Siemonsen), **Summer School on Gravitational Wave Astronomy**, ICTS-TIFR, Bengaluru.

Oct 2023 Tutor for Prof. Badri Krishnan's course on Dynamical Horizons, **IAGRG School on Gravitation and Cosmology**, ICTS-TIFR, Bengaluru.

Jul 2023 Tutor for Prof. Harald Pfeiffer's course on pseudo-spectral methods, SpECTRE and numerical relativity, **Summer School on Gravitational-Wave Astronomy**, ICTS-TIFR, Bengaluru.

Jul 2023 Tutor for Prof. Geoffrey Lovelace's course on Numerical Relativity, **Summer School on Gravitational-Wave Astronomy**, ICTS-TIFR, Bengaluru.

Academic Service

Member, *LIGO-Virgo-KAGRA* Collaboration and *Simulating eXtreme Spacetimes* (SXS) Collaboration.

International Gravitational-Wave Observatory Network (IGWN): pledged time to build a surrogate model for precessing binaries.

Jul 2026 Co-organizer of the discussion session on *Testing General Relativity*, New Frontiers in Strong Gravity, Benasque Science Center, Spain.

2026 Referee for *Living Reviews in Relativity* (two review rounds).

Jan 2025 - Co-lead of the *Cosmic Explorer Science Traceability Matrix - Testing General Relativity*.

May 2025 Organizer of a discussion meeting on *Cosmic Explorer Science Traceability Matrix* at PAX Group, Penn State.

Jul 2024 Liaison for one of the courses in the *Summer School on Gravitational Wave Astronomy* at ICTS-TIFR.

Feb 2020 Organizing team, *IUCAA Annual Student Seminars*.

2018 Founding member and organizing team, *Informal Tuesday Seminars*, IUCAA.

Organizing and hosting team, *Visitors to IUCAA Gravitational Wave Research Group*.

Founding member, Green initiatives at IUCAA.

Science outreach activities

Aug 2023	Host for the *Chandrayaan-3 live soft-landing* event at Brigade Panorama housing society, Bengaluru, 23 August 2023.
Sep 2020	Public question-and-answer session on LIGO outreach activities, *LIGO tour with Minecraft*, 14 September 2020.
Dec 2019	Short video clip on *Types of Eclipses* on the occasion of the annular solar eclipse, IUCAA Science Centre, 26 December 2019.
Aug 2019	Vigyan Samagam 2019, Bengaluru, India. Popular science talk on ***Gravitational Waves*** for school children, delivered in the local language Kannada.
Nov 2018	*Q&A with a scientist*, a public discussion at the IUCAA Pulastya Science Festival, IUCAA Science Centre, 26 November 2018.

National Science Day (NSD) activities at the Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India.

1. Poster presentation on stellar formation and evolution, NSD 2016, 17, 19.
2. Quiz master for the science quiz, NSD 2018.
3. Organizing team for the NSD quiz, 2018, 19.

Computing skills and interests

Prog. lang.	Python(A), Mathematica, MATLAB, shell/bash, C, C++, FORTRAN
GW Data An.	pycbc, lalsuite, bilby, Bayesian inference.
Comp. Phy.	Data Modelling and Statistics, Numerical methods, Numerical solutions of differential equations.
HPC	Parallel Computing and Data I/O, MPI, GPU/CUDA, PBS/Torque, LSF, HTCondor, SLURM.
OS	Linux/Unix, Windows.
Other	Networking, Computer hardware and software
Utilities	Anaconda, Git, vi, Jupyter Notebook

Codes developed

tdanalysis	An accelerated, GPU-enabled exact time-domain inference engine for gravitational wave astronomy, using the Gohberg–Semencul representation of Toeplitz inverses.
nrhjsurrogate	A GPU-native numerical relativity surrogate waveform family providing analytic derivatives and error estimates. Released on PyPI.
spectral/spectools	A collection of spectral transforms and helpers with built-in error monitoring. Released on PyPI.
paralleltemplatebank	A massively parallel reduced-spin template bank generator for precessing compact binaries with higher modes.
purohit	A library to semi-automate gravitational wave parameter estimation.
popsampler	A sampler for drawing events from GWTC population models.
sxstools	Tools for handling SXS numerical relativity catalogue data.
gwdetectors	A library of gravitational wave detector models and sensitivity curves.
waveformtools	A library of data handling functions for the Einstein Toolkit and SpEC. Released on PyPI.
qlmtools	A library of functions for analysis and handling of 2D dynamical horizon numerical data.
quasinormal	A library of functions for analysis of quasi-normal modes.
wfs_match	An MPI-based parallel template parameter estimation routine.
yumcee	An MPI-based Markov chain and nested Monte Carlo sampler.
null_solver	An implementation of the dual-null characteristic numerical solver for scattering problems.
pkginstaller	A simple package installer for HPCs.
nrcatalogtools	A unified interface to numerical relativity waveform catalogs. Contributor.

Communication

Foreign English (A, SRW), French (I, SRW), German (B, R)

Indian Kannada (A, SRW), Sanskrit (I, SRW), Hindi (I, SRW)

A: advanced, I: intermediate, B: basic. S: speak, R: read, W: write.

Conferences and Schools

Jul 2026 New Frontiers in Strong Gravity, Benasque Science Center, Benasque, Spain Co-organizer of the discussion session on Testing General Relativity

Jul 2026 Horizons, Singularities and Censorship in Gravitational Collapse, CENTRA, Instituto Superior Técnico, Lisbon, Portugal

Jun 2026 Eastern Gravity Meeting 2026, Cornell University, Ithaca, NY.

Mar 2026 APS Global Physics Summit 2026, Denver, CO Invited speaker

Nov 2025 APS Mid Atlantic Section 2025, Penn State Harrisburg

Oct 2025 Workshop on Big Data and AI, Pittsburgh Supercomputing Center, PA.

Sep 2025 deciHz on the Moon, Johns Hopkins University, Baltimore, MD.

Jun-Jul 2025 PAX-X and CE Symposium, University of Illinois Urbana-Champaign, IL. Panellist

Jun 2025 NP3M Summer School, Indiana University, IN. Tutor/lecturer

Jun 2025 Scientific Machine Learning for Gravitational Wave Astronomy, ICERM, Providence, RI

May 2025 Science Traceability Matrix for Cosmic Explorer, PSU, State College, PA. Organizer

Apr 2025 APS Global Summit, Anaheim, CA

Aug 2024 Black Holes Inside Out, NBI, Copenhagen, Denmark

Aug 2024 Ringdown Inside Out, NBI, Copenhagen, Denmark

Aug 2024 Simulating Extreme Spacetimes - Conference and Developer Meeting 2024, ICERM, Providence, RI, USA

Mar 2024 The RIKEN Spring School on High Performance Computing - Krylov Subspaces, Finite Element Methods and Parallel Computing, R-CCS, Kobe, Japan

Dec 2023 The 10th International Conference on Gravitation and Cosmology (ICGC), IIT Guwahati

Feb 2022 Second Chennai Symposium on Gravitation and Cosmology

Jan 2022 School on Black Holes and Gravitational Waves, Centre for Strings, Gravitation and Cosmology (CSGC), Indian Institute of Technology Madras (IITM), Chennai, India

Jul 2021 Amaldi 14, (Virtual Conference).

Aug 2020 The North American Einstein Toolkit workshop.

Dec 2019 International Conference on Gravitation and Cosmology (ICGC), Indian Institute of Science Education and Research, Mohali (IISER-Mohali), India.

Sep 2019 The European Einstein Toolkit workshop, King's College London, United Kingdom (Remote Attendee).

Jan 2019 30th meeting of the Indian Association for General Relativity and Gravitation (IAGRG), Birla Institute of Technology and Science, Pilani - Hyderabad Campus, Hyderabad, India.

Aug 2018 The fourth meeting on the Physics and Astrophysics at the eXtreme (PAX-IV), Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India.

Aug 2018 Summer School on Gravitational-Wave Astronomy, International Centre for Theoretical Sciences (ICTS), Bengaluru, India.

Jul 2018 The Les Houches Summer School on Gravitational Waves, École de physique des Houches, Les Houches, France.

Sep 2017 Young Astronomers Meet, Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India.

References

Letters of reference available upon request.